

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and developed new user-facing features for an enterprise Laboratory Information System, focusing on **enhanced functionality** and **continuous improvement**.
- Managed and improved the transfer of data between laboratory instruments, handling **ever-increasing volumes of data throughput** and implementing **data quarantine systems**.
- Developed back-end web applications and created/modified databases to support new functionalities, optimizing applications for **ease of use, speed, and scalability**.
- Collaborated with cross-functional teams, including analytical, field services, quality, finance, and sales, to understand and implement their **data requirements** and automate business processes.
- Designed and developed APIs, building reusable code and libraries for future use, and ensured applications adhered to **script procedures** and logical code layout.
- Troubleshoot and resolved issues within on-premise and cloud environments, contributing to automated quality control verifications and **data management tasks**.
- Containerized applications using **Docker** and deployed them to **AWS EC2** instances, managing CI/CD pipelines with **Jenkins** for streamlined releases.

SKILLS

Languages – PHP, Python, SQL, Java, R

Web Technologies – REST APIs, Web Applications, HTML, CSS

Databases – PostgreSQL, MySQL, Oracle, MS SQL Server

Cloud – AWS (EC2, S3, Lambda, CloudWatch), Cloud Computing Infrastructure

Development Tools – Docker, Jenkins, Git, Jira, Confluence, Postman

Analytics & Data Management – Data Computations, Data Transfer, Data Quarantine, Statistical Analysis, Method Validations, Bias, Precision, Uncertainty of Measurement Controls, MS-Office Suite, Power BI, Tableau, DAX, Power Query, pandas, NumPy, scikit-learn

Methodologies – Agile, Scrum, Change Management, Release Deployment Documentation

PROJECTS

Laboratory Data Integration Platform

- Developed a Python-based data integration service to manage data flow from laboratory instruments to a central database, processing approximately **15,000 data points daily**.
- Implemented automated quality control checks and data quarantine mechanisms to ensure data integrity and compliance with environmental monitoring standards.
- Designed and built REST APIs using **Flask** for external systems to access processed laboratory data, supporting integration with downstream reporting tools.

Instrument Data Throughput Optimization

- Engineered a Java Spring Boot application to optimize data processing pipelines, significantly improving the handling of large data volumes from analytical instruments.
- Refactored database schemas in **PostgreSQL** to accommodate increased data throughput and implement efficient indexing, reducing query times by an average of **30%**.
- Collaborated with the operations team to define requirements for data transfer improvements, ensuring seamless integration with existing laboratory workflows.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)